

Linux File Management Machine Test

Objectives

- 1 Create an empty file called x in the /tmp directory.
- 2 Copy the x file from /tmp to your home directory.
- 3 Create a copy of the x file and name it y.
- 4 Create a directory called files and move both x and y into it.
- 5 Create a copy of the files directory and name the copy copy_of_files.
- 6 Rename the copy_of_files directory to files2.
- 7 Remove both files and files2 directories along with their contents.

```
anish@rakusafe: ~  
anish@rakusafe:~$  
anish@rakusafe:~$ sudo touch /tmp/x  
anish@rakusafe:~$ ls  
DevOps-Team  ani.txt      deploy.sh    linux-test  
DevOpsLab    application.log 'deploy.sh' '$'\341\271\243' logs.txt  
LinuxPractice backup       developer    notes.txt  
MyLinuxLab   config.env   employee.txt project.txt  
ani          config.txt   home        x  
anish@rakusafe:~$ cd /tmp/  
anish@rakusafe:/tmp$ ls  
snap-private-tmp  
systemd-private-939b218340ed4ad28cae0e9e57df88f4-chrony.service-UJvFif  
systemd-private-939b218340ed4ad28cae0e9e57df88f4-systemd-logind.service-EWJZdF  
x  
anish@rakusafe:/tmp$ cd  
anish@rakusafe:~$ sudo cp /tmp/x ~/x  
anish@rakusafe:~$ ls  
DevOps-Team  ani.txt      deploy.sh    linux-test  
DevOpsLab    application.log 'deploy.sh' '$'\341\271\243' logs.txt  
LinuxPractice backup       developer    notes.txt  
MyLinuxLab   config.env   employee.txt project.txt  
ani          config.txt   home        x  
anish@rakusafe:~$ mv touch
```

```
anish@rakusafe: ~/files2
anish@rakusafe:~$
anish@rakusafe:~$
anish@rakusafe:~$ sudo mkdir files
anish@rakusafe:~$ mv x y files/
mv: cannot move 'x' to 'files/x': Permission denied
mv: cannot move 'y' to 'files/y': Permission denied
anish@rakusafe:~$ sudo mv x y files/
anish@rakusafe:~$ ls
DevOps-Team  ani.txt      deploy.sh    home
DevOpsLab    application.log 'deploy.sh'$'\341\271\243' linux-test
LinuxPractice backup       developer    logs.txt
MyLinuxLab   config.env   employee.txt notes.txt
ani          config.txt   files        project.txt
anish@rakusafe:~$ ls -l files/
total 0
-rw-r--r-- 1 root  root  0 Oct  4 13:13 x
-rw-r--r-- 1 anish anish 0 Oct  4 13:23 y
```

```
anish@rakusafe:~$ sudo mv files files2
anish@rakusafe:~$ ls
DevOps-Team  ani.txt      deploy.sh    home
DevOpsLab    application.log 'deploy.sh'$'\341\271\243' linux-test
LinuxPractice backup       developer    logs.txt
MyLinuxLab   config.env   employee.txt notes.txt
ani          config.txt   files2       project.txt
anish@rakusafe:~$ cd files2/
anish@rakusafe:~/files2$ ls
x y
```

```
anish@rakusafe: ~  
anish@rakusafe:~$ sudo ls -l files/  
total 0  
-rw-r--r-- 1 root root 0 Oct 4 13:13 x  
-rw-r--r-- 1 anish anish 0 Oct 4 13:23 y  
anish@rakusafe:~$ sudo cp -r files copy_of_files  
anish@rakusafe:~$ ls  
DevOps-Team    application.log  'deploy.sh'$'\341\271\243'  logs.txt  
DevOpsLab      backup          developer          notes.txt  
LinuxPractice  config.env      employee.txt       project.txt  
MyLinuxLab     config.txt      files  
ani            copy_of_files  home  
ani.txt        deploy.sh      linux-test  
anish@rakusafe:~$ sudo ls -l files/  
total 0  
-rw-r--r-- 1 root root 0 Oct 4 13:13 x  
-rw-r--r-- 1 anish anish 0 Oct 4 13:23 y  
anish@rakusafe:~$ sudo ls -l co  
config.env      config.txt      copy_of_files/  
anish@rakusafe:~$ sudo ls -l copy_of_files/  
total 0  
-rw-r--r-- 1 root root 0 Oct 4 14:05 x  
-rw-r--r-- 1 root root 0 Oct 4 14:05 y  
anish@rakusafe:~$ sudo mv co  
config.env      config.txt      copy_of_files/  
anish@rakusafe:~$ sudo mv copy_of_files/ files2  
anish@rakusafe:~$ ls  
DevOps-Team    application.log  developer          logs.txt  
DevOpsLab      backup          employee.txt       notes.txt  
LinuxPractice  config.env      files              project.txt  
MyLinuxLab     config.txt      files2  
ani            deploy.sh       home  
ani.txt        'deploy.sh'$'\341\271\243'  linux-test  
anish@rakusafe:~$ sudo ls -l files2/  
total 0  
-rw-r--r-- 1 root root 0 Oct 4 14:05 x  
-rw-r--r-- 1 root root 0 Oct 4 14:05 y  
anish@rakusafe:~$ rm -r files  
files/ files2/  
anish@rakusafe:~$ rm -r files/ files2/  
rm: descend into write-protected directory 'files/'?
```

Linux File and Directory Management Machine Test

Objectives

1. Create a file called **x**.
2. Create a directory called **content**.
3. Move the **x** file to the **content** directory.
4. Create a file inside the **content** directory called **y**.
5. Create the following directory structure inside the **content** directory: **dir1/dir2/dir3**.
6. Remove the **content** directory along with all its contents.

```
anish@rakusafe: ~
anish@rakusafe:~$
anish@rakusafe:~$ sudo touch x
anish@rakusafe:~$
anish@rakusafe:~$ sudo mkdir content
anish@rakusafe:~$ ls
DevOps-Team  ani.txt          content          home            x
DevOpsLab    application.log  deploy.sh       linux-test
LinuxPractice backup           'deploy.sh'$'\341\271\243' logs.txt
MyLinuxLab   config.env       developer        notes.txt
ani          config.txt       employee.txt     project.txt
anish@rakusafe:~$ sudo mv x con
config.env config.txt content/
anish@rakusafe:~$ sudo mv x content/
anish@rakusafe:~$ ls
DevOps-Team  ani.txt          content          home            linux-test
DevOpsLab    application.log  deploy.sh       'deploy.sh'$'\341\271\243' logs.txt
LinuxPractice backup           developer        notes.txt
MyLinuxLab   config.env       employee.txt     project.txt
ani          config.txt
anish@rakusafe:~$ sudo ls -l con
config.env config.txt content/
anish@rakusafe:~$ sudo ls -l content/
total 0
-rw-r--r-- 1 root root 0 Oct  4 14:18 x
anish@rakusafe:~$ sudo touch con
config.env config.txt content/
anish@rakusafe:~$ sudo touch content/y
anish@rakusafe:~$ ls
DevOps-Team  ani.txt          content          home            linux-test
DevOpsLab    application.log  deploy.sh       'deploy.sh'$'\341\271\243' logs.txt
LinuxPractice backup           developer        notes.txt
MyLinuxLab   config.env       employee.txt     project.txt
ani          config.txt
anish@rakusafe:~$ sudo ls -l content/
total 0
-rw-r--r-- 1 root root 0 Oct  4 14:18 x
-rw-r--r-- 1 root root 0 Oct  4 14:22 y
anish@rakusafe:~$
```

```
anish@rakusafe:~$ sudo mkdir -p content/dir1/dir2/dir3
anish@rakusafe:~$ ls
DevOps-Team    ani.txt        content        home
DevOpsLab      application.log deploy.sh       linux-test
LinuxPractice  backup        'deploy.sh'$'\341\271\243' logs.txt
MyLinuxLab     config.env     developer      notes.txt
ani           config.txt     employee.txt   project.txt
anish@rakusafe:~$ cd content
anish@rakusafe:~/content$ tree
.
├── dir1
│   └── dir2
│       └── dir3
├── x
└── y
4 directories, 2 files
anish@rakusafe:~/content$
```

```
anish@rakusafe: ~
anish@rakusafe:~$
anish@rakusafe:~$
anish@rakusafe:~$
anish@rakusafe:~$
anish@rakusafe:~$
anish@rakusafe:~$
anish@rakusafe:~$
anish@rakusafe:~$ rm -r con
config.env config.txt content/
anish@rakusafe:~$ rm -r content/
rm: descend into write-protected directory 'content/'? ^C
anish@rakusafe:~$ sudo rm -r content/
anish@rakusafe:~$
anish@rakusafe:~$ ls
DevOps-Team    ani.txt        deploy.sh       linux-test
DevOpsLab      application.log 'deploy.sh'$'\341\271\243' logs.txt
LinuxPractice  backup        developer      notes.txt
MyLinuxLab     config.env     employee.txt   project.txt
ani           config.txt     home
```

Machine Test 03 – File Permissions

Objectives

1. Create a file called test.txt.
2. Display the current permissions of test.txt.
3. Change the permissions of test.txt so that:
 - Owner has read and write permissions.
 - Group has read-only permission.
 - Others have read-only permission.
4. Verify the updated permissions of test.txt.
5. Create a shell script called deploy.sh.
6. Grant executable permission to deploy.sh.
7. Verify that deploy.sh is executable.
8. Create a directory called secure_data.
9. Assign the following permissions to secure_data:
 - Owner has full access.
 - Group has read and execute permissions.
 - Others have no permissions.
10. Verify the permissions of secure_data.
11. Remove test.txt, deploy.sh, and secure_data.

```
anish@rakusafe: ~  
anish@rakusafe:~$  
anish@rakusafe:~$ sudo touch test.txt  
[sudo: authenticate] Password:  
sudo: Authentication failed, try again.  
[sudo: authenticate] Password:  
anish@rakusafe:~$  
anish@rakusafe:~$  
anish@rakusafe:~$ ls  
DevOps-Team  ani.txt      deploy.sh      linux-test  
DevOpsLab    application.log 'deploy.sh'$'\341\271\243' logs.txt  
LinuxPractice backup        developer       notes.txt  
MyLinuxLab   config.env    employee.txt    project.txt  
ani          config.txt    home            test.txt  
anish@rakusafe:~$ sudo ls -ld test.txt  
-rw-r--r-- 1 root root 0 Oct 5 09:06 test.txt  
anish@rakusafe:~$ sudo touch deploy.sh  
anish@rakusafe:~$ ls  
DevOps-Team  ani.txt      deploy.sh      linux-test  
DevOpsLab    application.log 'deploy.sh'$'\341\271\243' logs.txt  
LinuxPractice backup        developer       notes.txt  
MyLinuxLab   config.env    employee.txt    project.txt  
ani          config.txt    home            test.txt  
anish@rakusafe:~$ ls -ld deploy.sh  
-rwxr-xr-x 1 root root 65 Oct 5 09:10 deploy.sh  
anish@rakusafe:~$ sudo mkdir secure_data  
anish@rakusafe:~$ ls  
DevOps-Team  application.log      developer  project.txt  
DevOpsLab    backup              employee.txt secure_data  
LinuxPractice config.env           home       test.txt  
MyLinuxLab   config.txt           linux-test  
ani          deploy.sh            logs.txt  
ani.txt      'deploy.sh'$'\341\271\243' notes.txt  
anish@rakusafe:~$ ls -ld secure_data/  
drwxr-xr-x 2 root root 4096 Oct 5 09:11 secure_data/  
anish@rakusafe:~$ sudo chmod 750 secure_data/
```

```
anish@rakusafe:~$ sudo rm -rf test.txt deploy.sh secure_data/  
anish@rakusafe:~$  
anish@rakusafe:~$ ls  
DevOps-Team  ani.txt      'deploy.sh'$'\341\271\243' logs.txt  
DevOpsLab    application.log developer     notes.txt  
LinuxPractice backup        employee.txt project.txt  
MyLinuxLab   config.env    home  
ani          config.txt    linux-test  
anish@rakusafe:~$
```

Machine Test 04 – Users and Groups

Objectives

1. Create a new user called devopsuser.
2. Verify that the user devopsuser has been created successfully.
3. Create a new group called devops.
4. Add devopsuser to the devops group.
5. Verify the group membership of devopsuser.
6. Create a directory called project_data.
7. Change the ownership of project_data to devopsuser.
8. Change the group ownership of project_data to devops.
9. Verify the ownership and permissions of project_data.
10. Create a file called project.txt inside project_data.
11. Assign ownership of project.txt to devopsuser and devops group.
12. Verify the ownership details of project.txt.
13. Delete the project.txt file.
14. Remove the project_data directory.
15. Delete the devopsuser user.
16. Delete the devops group.

```
anish@rakusafe: ~  
anish@rakusafe:~$ sudo useradd devopsuser  
anish@rakusafe:~$ cat /etc/passwd | grep devopsuser  
devopsuser:x:1001:1008:./home/devopsuser:/bin/sh  
anish@rakusafe:~$ sudo groupadd devops  
groupadd: group 'devops' already exists
```

```
Select anish@rakusafe: ~  
anish@rakusafe:~$  
anish@rakusafe:~$ sudo usermod -aG dev  
developers devops devopsuser  
anish@rakusafe:~$ sudo usermod -aG devops devopsuser  
anish@rakusafe:~$ sudo cat /etc/passwd | grep de  
deploy.sh$ developer/  
anish@rakusafe:~$ sudo cat /etc/passwd | grep de  
deploy.sh$ developer/  
anish@rakusafe:~$ sudo cat /etc/passwd | grep devopsuser  
devopsuser:x:1001:1008:./home/devopsuser:/bin/sh  
anish@rakusafe:~$ sudo cat /etc/group | grep devops  
devops:x:1007:devopsuser  
devopsuser:x:1008:  
anish@rakusafe:~$ mkdir project_data  
anish@rakusafe:~$ sudo chown  
_apt dhcpcd man sync  
_chrony games messagebus sys  
anish irc news syslog  
backup landscape nobody systemd-network  
bin list polkitd systemd-resolve  
daemon lp proxy uucp  
devopsuser mail root www-data  
anish@rakusafe:~$ sudo chown devopsuser project  
project.txt project_data/  
anish@rakusafe:~$ sudo chown devopsuser project  
project.txt project_data/  
anish@rakusafe:~$ sudo chown devopsuser project_data/  
anish@rakusafe:~$ sudo ls -ld project_data/  
drwxr-xr-x 2 devopsuser anish 4096 Oct 5 13:00 project_data/  
anish@rakusafe:~$ sudo chown devopsuser:devops project_data/  
anish@rakusafe:~$  
anish@rakusafe:~$ sudo ls -ld project_data/  
drwxr-xr-x 2 devopsuser devops 4096 Oct 5 13:00 project_data/
```

```
anish@rakusafe: ~  
anish@rakusafe:~$  
anish@rakusafe:~$  
anish@rakusafe:~$ sudo touch project_data/project.txt  
anish@rakusafe:~$ sudo ls -ld project  
project.txt  project_data/  
anish@rakusafe:~$ sudo ls -ld project_data/  
drwxr-xr-x 2 devopsuser devops 4096 Oct  5 13:25 project_data/  
anish@rakusafe:~$ sudo ls -ld project_data/project.txt  
-rw-r--r-- 1 root root 0 Oct  5 13:25 project_data/project.txt  
anish@rakusafe:~$ sudo chown devopsuser:dev  
developers devops devopsuser  
anish@rakusafe:~$ sudo chown devopsuser:devops project.txt  
^[[6~^[[6~anish@rakusafe$ sudo ls -ld project_data/project.txt  
-rw-r--r-- 1 root root 0 Oct  5 13:25 project_data/project.txt  
anish@rakusafe:~$ ^C  
anish@rakusafe:~$ sudo chown devopsuser:devops project_data/project.txt  
anish@rakusafe:~$ sudo ls -ld project_data/project.txt  
-rw-r--r-- 1 devopsuser devops 0 Oct  5 13:25 project_data/project.txt  
anish@rakusafe:~$
```

```
anish@rakusafe:~$ sudo chown devopsuser:devops project_data/project.txt  
anish@rakusafe:~$ sudo ls -ld project_data/project.txt  
-rw-r--r-- 1 devopsuser devops 0 Oct  5 13:25 project_data/project.txt  
anish@rakusafe:~$ sudo rm -r project_data/  
anish@rakusafe:~$ sudo ls -ld project_data  
ls: cannot access 'project_data': No such file or directory
```

```
anish@rakusafe:~$ sudo userdel devopsuser  
anish@rakusafe:~$ sudo cat /etc/passwd | grep devopsuser  
anish@rakusafe:~$ sudo groupdel dev  
developers devops  
anish@rakusafe:~$ sudo groupdel devops  
anish@rakusafe:~$ sudo cat /etc/group | grep devops  
anish@rakusafe:~$
```